Peer Review 1

Graded

19 Hours, 28 Minutes Late

Student

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Total Points

1 / 1 pts

Question 1

Peer Review 1 **1** / 1 pt

✔ **+ 1 pt** Correct

+ 0.75 pts Not a substantitive review

– 0 pts Incomplete

Question assigned to the following page: 1

Multi-agent Framework for Among Us

What is the author trying to do?
In this video, the authors aim to construct a multi-agent framework for playing the hit game Among Us. The agent is focused on a crewmate's perspective, with the goal of identifying imposters based on other players' behavioral cues, position data, and chat logs during crewmate meetings in-game. The framework combines analysis of live human gameplay with scraped gameplay metrics in offline-to-online reinforcement learning.

Is the goal of this paper significant?
Among Us is a perfect game to model partial observability, as the state of the imposters' identities is unknown to the player, and it requires an agent to be robust against "deception" by the imposters as well. It mirrors how an agent can analyze real human behavior from limited, noisy observations to form conclusions about social behavior (e.g. motives). The discussion of previous work of Tim Shaw on a ChatGPT-based chat analyzer is relevant and helps to frame this project within the field of Among Us predictability.

Is the method of approach valid?
Given the limited data and difficulty of training a full POMDP model, approximating the environment as an MDP with many dimensions and alternatively-engineered history rather than a large event horizon is a reasonable choice. It is a strong idea that the reward has higher penalties for shrinking lobbies (aka. more non-imposters getting killed in a state that benefits the imposters), and that colors are swapped to force the agent to prioritize behavior over color. While data augmentation is valid, the video is unclear on how modifying "player permutations" extends 500 games to 15,000. A more explicit explanation with diagrams or formulas may be helpful, but this makes sense given the length limitation of the video. The reliance on human-engineered history limits how well it actually models true history.

Is the actual execution of the research correct?
The execution is well thought out. They explain how the data is collected and parsed at a high level. Also, the reward function for voting decisions makes sense for the crewmate's goals of efficiently voting out the imposter before other crewmates do, and doing so correctly. Building an action mask based on the number of dead crewmates and itself makes sense too. However, with 229 states and 19 actions, the problem is rather complex relative to the number of true games (500). The authors are aware of these limitations, though, and they seem to do a good job at designing around it. However, it raises the question of whether the agent is learning from the games or exploiting side-effects that arise due to manually augmenting these games.

Are the correct conclusions being drawn from the results?

Question assigned to the following page: 1

They correctly acknowledge that human baselines are hard to obtain, as each player's behavior varies. Using "voting out the imposter before the majority" makes sense as an evaluation metric as well. Also, the authors properly avoid overstating the statistical significance of their model given the limited data, which strengthens their conclusions/analysis. It would be fascinating to explore the effects of varying the hyperparameters in the reward function, such as making the agent more aggressive in suspecting imposters or more conservative to prevent lobbies from shrinking.

Is the presentation satisfactory?
The discussion of limitations and future work is strong. The authors are honest about the constraints, which is a large strength. For future work, extending the model to incorporate actual state history is a good next step, and improving chat analysis with an LLM rather than just keyword analysis will only make it more robust to the large variation in chat messages across the internet. However, some aspects remain unclear, likely due to time constraints of having a video-only presentation. While the approach is clearly stated, the video is only somewhat clear about what all of these metrics (player position, not doing any actions when voting, etc.) mean, even for someone who has played Among Us (maybe in recent years it has changed). Having additional visuals and diagrams in a short paper or appendix would improve clarity in the future. In the end, this project does a great job in dealing with the complexity of human interactions, and is extremely relevant (despite being just a video game).